

Siyu Xie

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EDUCATION

The University of Chicago

Master of Science in Financial Mathematics

Chicago, IL

Expected December 2027

- **Upcoming:** Probability & Stochastic Processes, Stochastic Calculus, Multivariate Statistics, Numerical Methods

University of Nottingham

Ningbo, China & Nottingham, UK

Bachelor of Science in Computer Science with Artificial Intelligence (2+2; GPA: 3.9/4.0)

June 2026

- **Courses:** Linear Algebra, Optimization, Discrete Math, Machine Learning, AI Algorithms, C++, Data Structures
- **Awards:** Gold Level, WorldQuant Challenge; Meritorious Winner (Top 6%), Mathematical Contest in Modeling

SKILLS

Technical: Python, C++, SQL, PyTorch, Pandas, NumPy, scikit-learn, Jupyter, Git, Linux

Research: Factor Research, Statistical Modeling, Market Microstructure, Portfolio Construction, Optimization

EXPERIENCE

Yingling Asset Management Company

Ningbo, China

Quantitative Researcher Intern

July 2025 – August 2025

- Built a rolling backtesting pipeline for private-fund NAV data (18 funds + cash, 337 weekly observations), integrating point-in-time data handling, transaction costs, and 11 allocation/risk-overlay variants
- Evaluated strategy performance over 204 out-of-sample test weeks; identified inverse-volatility allocation as the most robust cost-aware specification, improving Sharpe from 1.59 to 4.23 raw and above 3.0 after NAV autocorrelation adjustment, while cutting max drawdown by 84.2%

Ningbo Bank

Ningbo, China

Architecture Intern

July 2024 – September 2024

- Fine-tuned Qwen1.5-14B with Firefly and built a financial RAG system using LlamaIndex, improving Q&A accuracy by 7.6% and supporting automated generation of 50+ monthly investment analysis reports

RESEARCH

University of Nottingham

Nottingham, UK

Reinforcement Learning Order Book Trading Agent

March 2026 – June 2026

- Constructed a PPO-based trading agent in a BSE simulated limit order book with ZIC/ZIP background traders, comparing five state representations across price-only, best bid/ask, raw LOB depth, handcrafted microstructure features, and learned LOB encodings
- Designed a risk-aware reward based on PnL changes, inventory exposure, and transaction costs; handcrafted microstructure states achieved 67% higher mean PnL than the best-bid/ask PPO baseline in denser-market validation ($p = 0.036$) and significantly outperformed three non-learning strategies ($p < 0.0001$)

Graph-Enhanced Reinforcement Learning Hyper-Heuristics for VRPTW

October 2025 – April 2026

- Designed a GraphSAGE-PPO hyper-heuristic for vehicle routing with time windows, selecting among 6 operators based on graph-based features to improve solution quality and cross-instance generalization

University of Nottingham Ningbo China

Ningbo, China

Budgeted Adaptive Falsification for Equity Alpha Discovery

February 2025 – August 2025

- Implemented a walk-forward alpha validation framework on a 499-stock U.S. equity universe (2010-2025 Tiingo data), evaluating 4,394 symbolic factors across 53,316 falsification tests and 14 temporal test episodes
- Developed a dynamic budget allocation policy to prune spurious signals, improving out-of-sample survivor precision by +3.1 pp and reducing false eliminations by 21.4 pp over a fixed-checklist baseline

Real-Time Trading Strategy System

June 2024 – November 2024

- Developed a PPO-LSTM trading agent with 24-step rolling market states and multi-source sentiment signals to model short-term market dynamics, achieving a Sharpe ratio of 2.69 on the DOGE test set under transaction costs
- Co-authored an IEEE CIEER 2025 paper on cryptocurrency trading strategies using deep reinforcement learning and multi-source LLM sentiment analysis; shortlisted for Best Paper Award

EXTRACURRICULAR

President, Computer Enthusiasts Association

September 2023 – September 2024

- Organized technical workshops, alumni Q&A sessions, and programming competitions for student members